



Autonomous SDR Engine: Executive Ghostwriting Pipeline



The Real-World Problem

B2B service providers, marketing agencies, and Executive Ghostwriters bleed hundreds of hours manually prospecting. The current standard operating procedure requires a human to:

1. Sift through raw lead databases (Apollo/ZoomInfo).
2. Manually open each prospect's LinkedIn profile.
3. Read through their recent posts to evaluate their current content quality.
4. Draft a custom cold outreach message based on those specific structural failures.

This is a highly inefficient, low-leverage task that drains revenue-generating time. It is a data-processing bottleneck that restricts scale.



The Solution: Dual-Agent Architecture

This project replaces the manual prospecting bottleneck with an event-driven, fully autonomous Sales Development Representative (SDR) built in n8n. The system ingests target leads from a CRM, autonomously scrapes and evaluates their public content, and drafts hyper-personalized outreach.

Rather than using AI as a glorified summarizer, this system utilizes a **multi-agent orchestration** to make executive decisions:

- **Agent 1 (The QA Bouncer):** Acts as a strict logic gate. It scans raw scraped social media posts and hunts for "POOR_FORMATTING" (e.g., walls of text, weak hooks, lack of structure). It filters out competent writers and only passes highly qualified leads to the next stage.
- **Agent 2 (The Pitch Architect):** Synthesizes the exact structural failures identified by Agent 1 to craft a bespoke, 2-sentence cold outreach pitch offering to rebuild their specific content architecture.



Technical Pipeline Breakdown

The n8n workflow executes the following automated sequence:

1. **Scheduled Ingestion (Trigger):** A cron job fires, pulling a batch of uncontacted leads from the CRM (Google Sheets synced with Apollo data).
2. **Rate-Limited Loop:** Leads are pushed through a Loop: Process 1-by-1 node containing a Wait throttle to prevent API bans and simulate human browsing velocity.

3. **Live Data Extraction:** An Apify headless browser node navigates to the prospect's LinkedIn URL and extracts their profile data and last 5 posts. (*Engineered with a 5-minute timeout protocol to guarantee execution stability*).
4. **AI Evaluation (Agent 1):** The LLM analyzes the JSON payload of posts to determine if the prospect needs ghostwriting services.
5. **Dynamic Synthesis (Agent 2):** If Agent 1 approves the lead, Agent 2 writes the hyper-personalized pitch.
6. **CRM Write-Back:** The completed pitch is logged directly adjacent to the lead in Google Sheets, and the lead is marked as "Processed" to prevent duplicate outreach.



Business Impact & ROI

- **Time to Value:** Converts a 10-hour weekly manual prospecting task into a continuous background process requiring zero human oversight.
- **Cost Efficiency:** Reduces the Customer Acquisition Cost (CAC) for outbound prospecting to the fraction of a cent required for API calls.
- **Conversion Optimization:** Replaces generic cold email templates with highly contextualized, proof-backed observations of the prospect's own content.